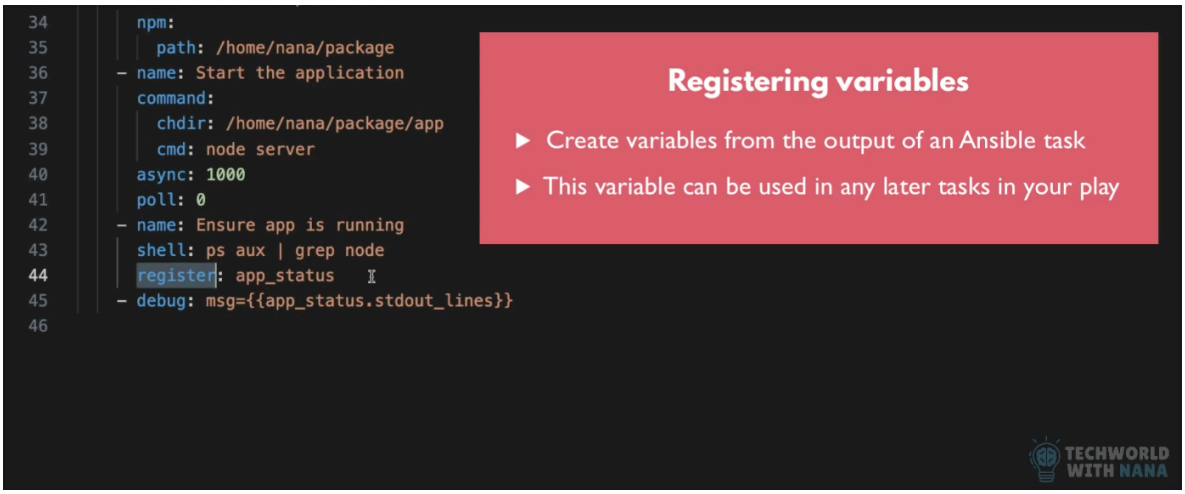
Module 15 - chapter 12

Ansible Variables - make your Playbook customizable

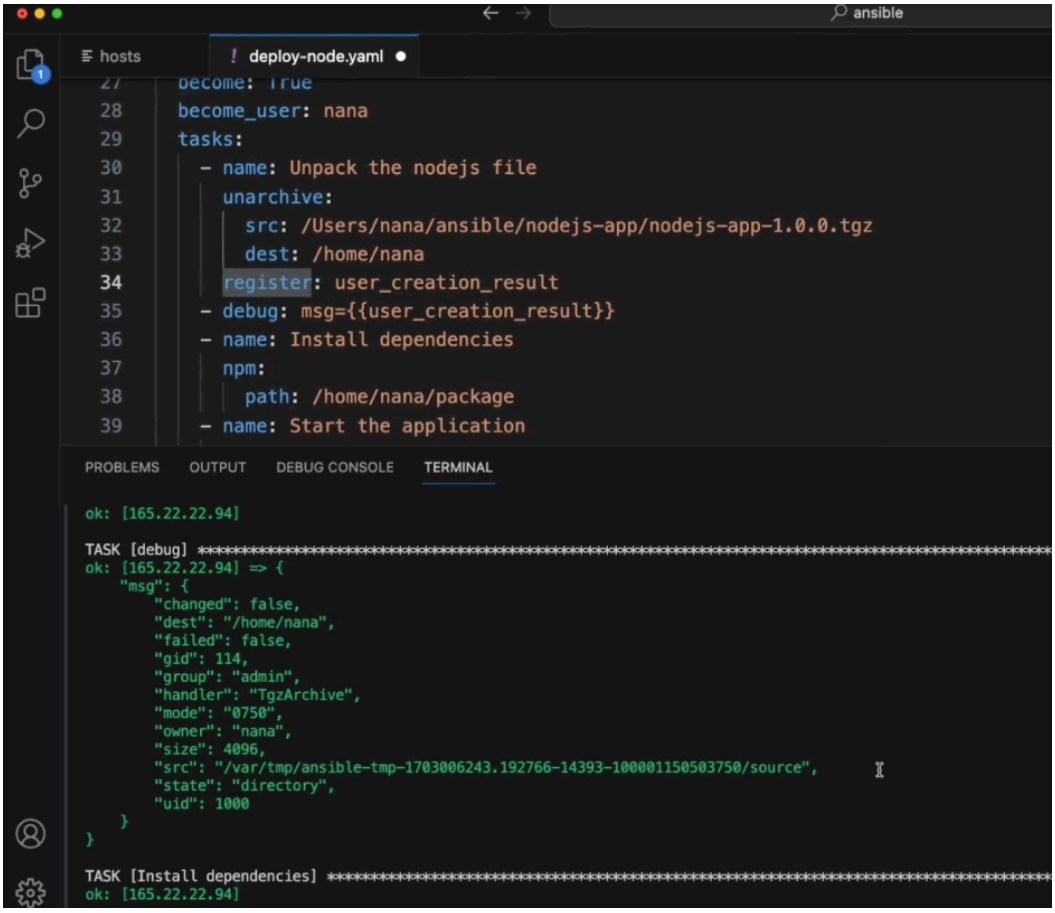
The project files used in this lecture can be found here: https://gitlab.com/twn-devops-bootcamp/latest/15-ansible/deploy-node-app/-/tree/feature/variables

Registered variables

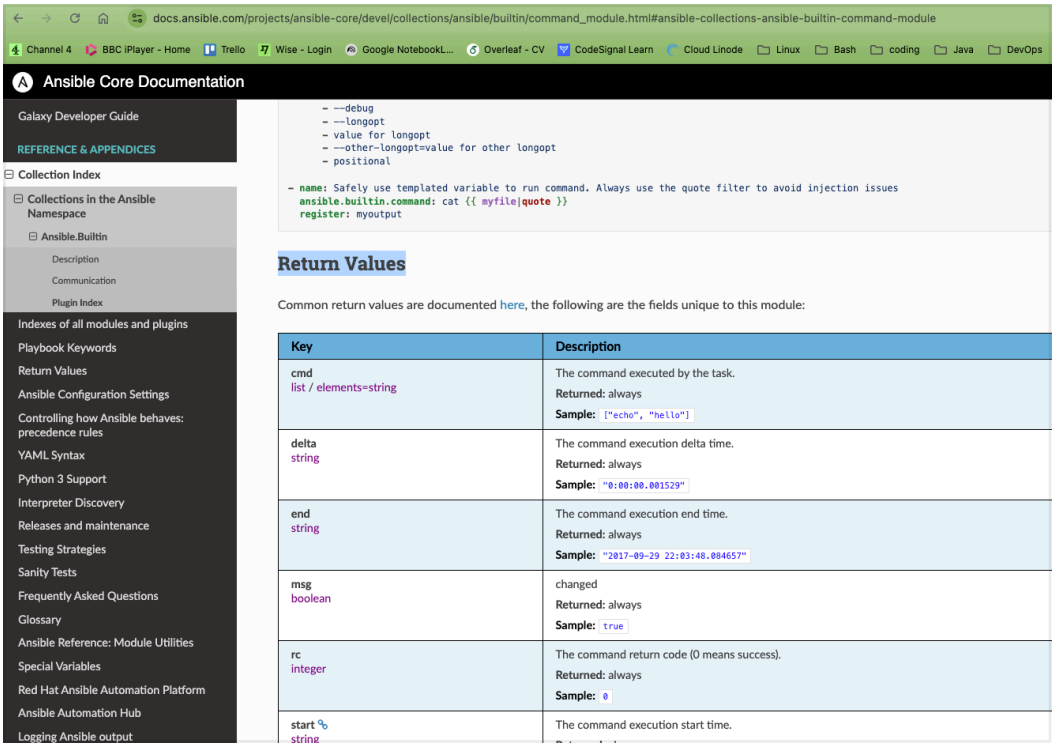
In the previous chapter (Module 15 - chapter 11) we used registered variables



We can get the results of other tasks for example the unarchive and use a registered variable to store the result of the task and then print it in th terminal using the ‘debug’ module

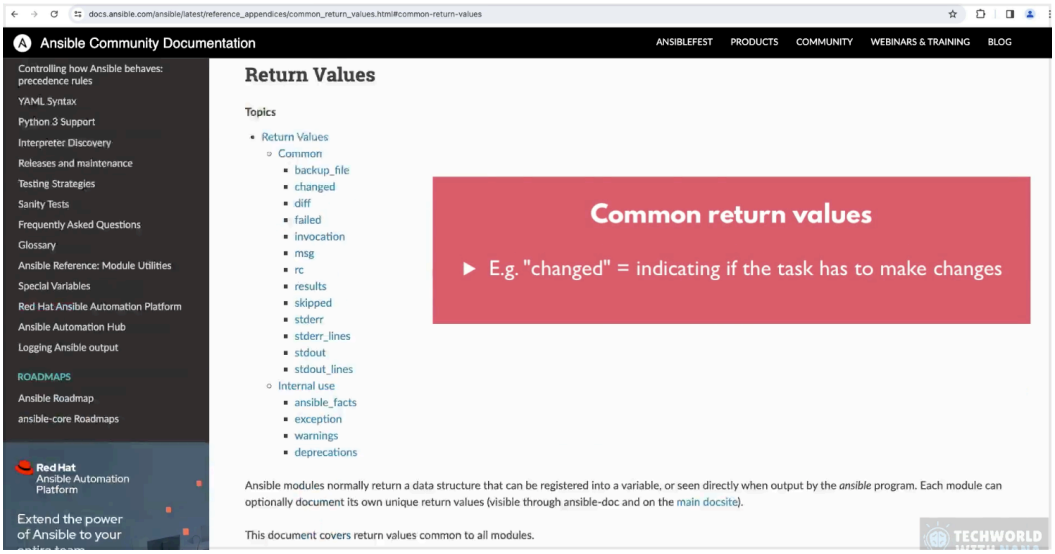


Not all of the modules will have documented the return information. If they do then we can see it in the documentation for example the ‘command’ module



And if we click on ‘here’ (in blue font in the screenshot above) then we go to

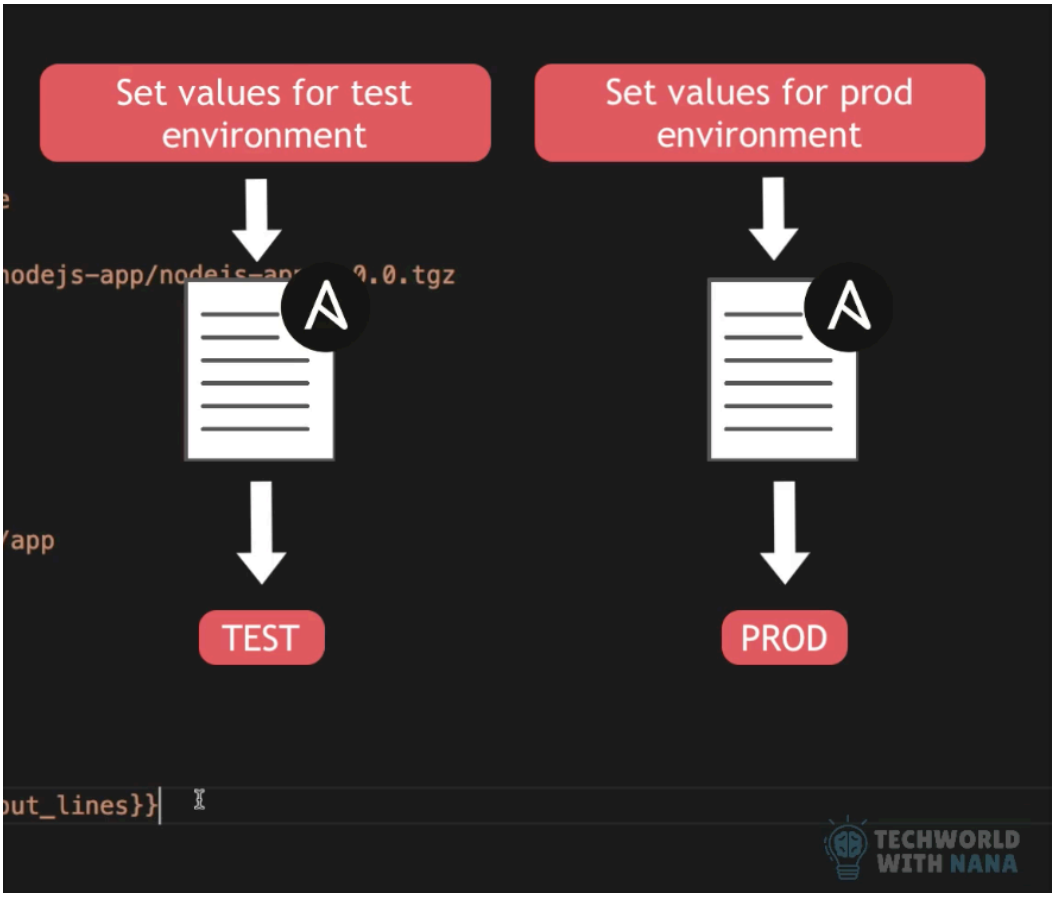
the list of Common return values:



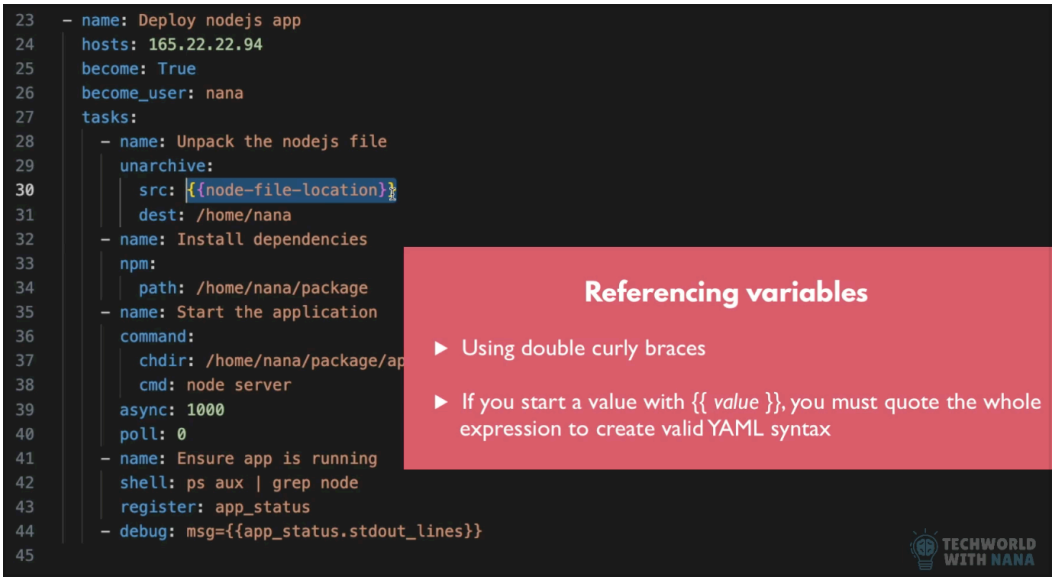
Nana suggest is better to print the whole result in the terminal and see what we get instead of checking them the documentation.

Parameterise Playbook

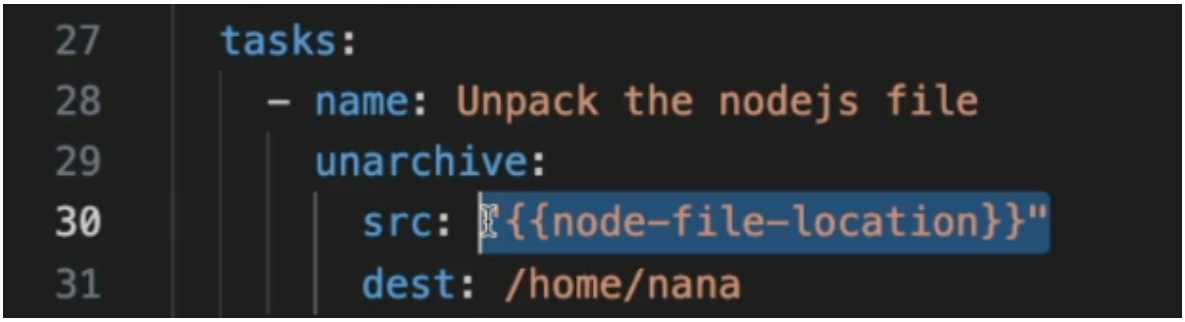
We want to make the values configurable per environment for example



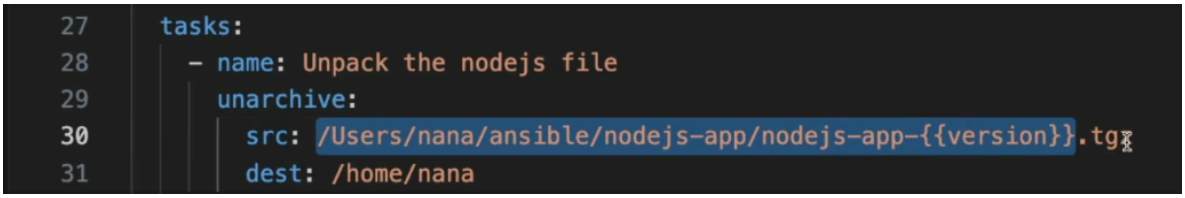
We will use double curly braces



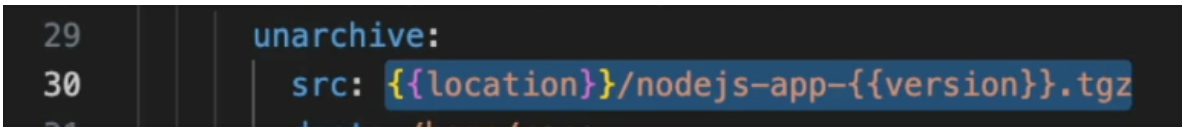
but if the curly braces go immediately after a colon we will need quotation marks, because ansible will think that it refers to a yaml dictionary syntax instead of a variable syntax.



If the curly braces are not immediately after a colon then we don't need the quotation marks:



If we need to parameterise part of the path



then we use “” but for the whole line

```
29        unarchive:
30          src: "{{location}}/nodejs-app-{{version}}.tgz"
```

This won't work:

```
29        unarchive:
30          src: "{{location}}"/nodejs-app-{{version}}.tgz
```

We can also parameterise '/home/astrid/' as it is repeated

```
26  tasks:
27    - name: Unpack the nodejs file
28      unarchive:
29        src: "{{location}}/nodejs-app/nodejs-app-{{version}}.tgz"
30        dest: /home/astrid
31    - name: Install dependencies
32      npm:
33        path: /home/astrid/package
34    - name: Start the application
35      command:
36        chdir: /home/astrid/package/app
37        cmd: node server
38      async: 1000
39      poll: 0
40    - name: Ensure app is running
41      shell: ps aux | grep node
42      register: app_status
43    - debug: msg={{app_status.stdout_lines}}
44
```

And now we have:

```yaml
tasks:
  - name: Unpack the nodejs file
    unarchive:
      src: "{{location}}/nodejs-app/nodejs-app-{{version}}.tgz"
      dest: "{{destination}}"
  - name: Install dependencies
    npm:
      path: "{{destination}}/package"
  - name: Start the application
    command:
      chdir: "{{destination}}/package/app"
      cmd: node server
    async: 1000
    poll: 0
  - name: Ensure app is running
    shell: ps aux | grep node
    register: app_status
  - debug: msg={{app_status.stdout_lines}}

```

Now we need to set the values, if we don't and execute the ansible playbook we will get an error
-> ansible-playbook -i hosts deploy-node.yaml

```
TASK [Unpack the nodejs file] ***************************************************************************************
[ERROR]: Task failed: Finalization of task args for 'ansible.builtin.unarchive' failed: Error while resolving value for 'dest':
 'destination' is undefined

Task failed.
Origin: /Users/astrid/PythonProject/Ansible/deploy-node.yaml:27:7

25   become_user: astrid
26   tasks:
27     - name: Unpack the nodejs file
         ^ column 7

<<< caused by >>>

Finalization of task args for 'ansible.builtin.unarchive' failed.
Origin: /Users/astrid/PythonProject/Ansible/deploy-node.yaml:28:7
```

Variables defined in a playbook

One way is to set them up under 'vast' attribute in the play and add the variables inside a dictionary structure:

```yaml
- name: Deploy nodejs app
  hosts: 46.101.49.110
  become: true
  become_user: astrid
  vars:

  tasks:
    - name: Unpack the nodejs file
      unarchive:
        src: "{{location}}/nodejs-app-{{version}}.tgz"
        dest: "{{destination}}"
    - name: Install dependencies
      npm:
        path: "{{destination}}/package"
    - name: Start the application
      command:
        chdir: "{{destination}}/package/app"
        cmd: node server
```

```yaml
- name: Deploy nodejs app
  hosts: 46.101.49.110
  become: true
  become_user: astrid
  vars: {
          location: /Users/astrid/PythonProject/Ansible/nodejs-app,
          version: 1.0.0,
          destination: /home/astrid
  }
  tasks:
    - name: Unpack the nodejs file
      unarchive:
        src: "{{location}}/nodejs-app-{{version}}.tgz"
        dest: "{{destination}}"
    - name: Install dependencies
      npm:
        path: "{{destination}}/package"
    - name: Start the application
      command:
        chdir: "{{destination}}/package/app"
        cmd: node server
```

And if I execute it
-> ansible-playbook -i hosts deploy-node.yaml

```
TASK [Ensure app is running] ***************************************************************************************
changed: [46.101.49.110]

TASK [debug] *******************************************************************************************************
ok: [46.101.49.110] => {
    "msg": [
        "astrid     41649 50.0  6.1 618048 60676 ?        Rl   14:29   0:00 node server",
        "astrid     41664  0.0  0.1   2800  1916 pts/1    S+   14:29   0:00 /bin/sh -c ps aux | grep node",
        "astrid     41666  0.0  0.2   7076  2164 pts/1    S+   14:29   0:00 grep node"
    ]
}

PLAY RECAP *********************************************************************************************************
46.101.49.110              : ok=11   changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

The second way is to make the playbook properly configurable instead of hardcoding the values inside the play.

Passing Variables On The command line

We will define the variables outside the playbook!

So let's say we want the location and the version to be set outside, so we removed them from 'vars'

```
22  - name: Deploy nodejs app
23    hosts: 46.101.49.110
24    become: true
25    become_user: astrid
26    vars: {
27          destination: /home/astrid
28    }
29    tasks:
30      - name: Unpack the nodejs file
31        unarchive:
```

and then we pass the values at the moment we run the command to execute the ansible playbook along with a flag '--extra-vars' or its shortcut '-e'
-> ansible-playbook -i hosts deploy-node.yaml --extra-vars

-> ansible-playbook -i hosts deploy-node.yaml -e

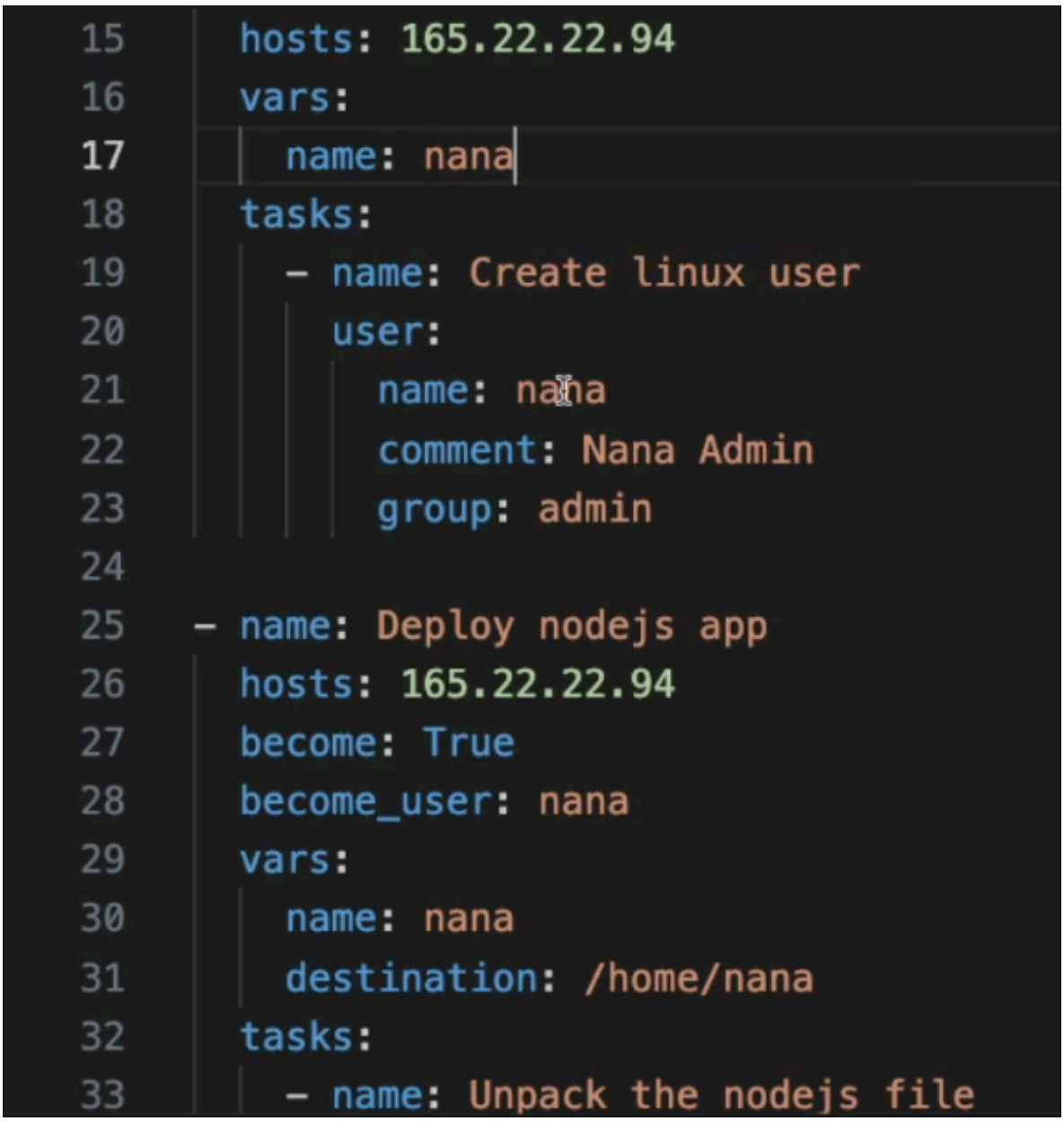
Along with the variables as a key:value pair
-> ansible-playbook -i hosts deploy-node.yaml -e "version=1.0.0 location=/Users/astrid/PythonProject/Ansible/nodejs-app"

```
❯ ansible-playbook -i hosts deploy-node.yaml -e "version=1.0.0 location=/Users/astrid/PythonProject/Ansible/nodejs-app"

PLAY [Install node and npm] ****************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [46.101.49.110]

TASK [Update apt repo and cache] ***********************************************************************************************
ok: [46.101.49.110]

TASK [Install nodejs and npm] **************************************************************************************************
ok: [46.101.49.110]

PLAY [Create new linux user for node app] **************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
```

```
ok: [46.101.49.110]

TASK [Create linux user] *******************************************************************************************************
ok: [46.101.49.110]

PLAY [Deploy nodejs app] *******************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [46.101.49.110]

TASK [Unpack the nodejs file] **************************************************************************************************
ok: [46.101.49.110]

TASK [Install dependencies] ****************************************************************************************************
ok: [46.101.49.110]
```

```
TASK [Start the application] ***************************************************************************************************
changed: [46.101.49.110]

TASK [Ensure app is running] ***************************************************************************************************
changed: [46.101.49.110]

TASK [debug] *******************************************************************************************************************
ok: [46.101.49.110] => {
    "msg": [
        "astrid     41649  0.1  6.0 618304 59280 ?        Sl   14:29   0:00 node server",
        "astrid     42249  0.0  0.1   2800  1912 pts/1    S+   14:39   0:00 /bin/sh -c ps aux | grep node",
        "astrid     42251  0.0  0.2   7076  2168 pts/1    S+   14:39   0:00 grep node"
    ]
}

PLAY RECAP *********************************************************************************************************************
46.101.49.110              : ok=11   changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

We want to use the user name ‘astrid’ and it is repeated in the current yaml. If we use the first way to set the value ‘astrid’ inside the playbook we will have to create add the ‘name’ in each play that we need it (See lines 17 and line 30)



This is not efficient, it is better to pass it as argument in the terminal.

So let's refactor:

```yaml
tasks:
  - name: Create linux user
    user:
      name: "{{name}}"
      comment: Astrid admin
      group: admin

- name: Deploy nodejs app
  hosts: 46.101.49.110
  become: true
  become_user: "{{name}}"
  vars: {
        destination: "/home/{{name}}"
  }
  tasks:
```

And now we have the variable ‘name’ but is not defined or hardcoded in each playbook.

Now we can execute the playbook and we need to pass the name as argument and actually let’s use ‘node’ as user instead of ‘astrid’
-> ansible-playbook -i hosts deploy-node.yaml -e "version=1.0.0 location=/Users/astrid/PythonProject/Ansible/nodejs-app name=node"

```
TASK [debug] ********************************************************************************************************************************
ok: [46.101.49.110] => {
    "msg": [
        "node       42935  0.0  2.1  40912 21072 ?        S    14:52   0:00 /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574
.9672909-14502-50420489012334/async_wrapper.py j221932103497 1000 false /var/tmp/ansible-tmp-1785941574.9672909-14502-504204896
12334/AnsiballZ_command.py /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574.9672909-14502-50420489012334/AnsiballZ_command.p
y",
        "node       42936  0.0  2.1  40912 21332 ?        S    14:52   0:00 /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574
.9672909-14502-50420489012334/async_wrapper.py j221932103497 1000 false /var/tmp/ansible-tmp-1785941574.9672909-14502-504204896
12334/AnsiballZ_command.py /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574.9672909-14502-50420489012334/AnsiballZ_command.p
y",
        "node       42937 22.2  2.9  40416 28584 ?        S    14:52   0:00 /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574
.9672909-14502-50420489012334/AnsiballZ_command.py",
        "node       42938 51.5  6.4 618056 63516 ?        Sl   14:52   0:00 node server",
        "root       42962  0.0  0.1   2800  1848 pts/1    Ss+  14:52   0:00 /bin/sh -c sudo -H -S -n  -u node /bin/sh -c 'echo
BECOME-SUCCESS-agvljtlylxktplmxqmzbohnbhvaummzr ; /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941576.5659258-14514-4260369505
684/AnsiballZ_command.py'",
        "root       42963  0.0  0.7  16948  7012 pts/1    S+   14:52   0:00 sudo -H -S -n -u node /bin/sh -c echo BECOME-SUCCES
S-agvljtlylxktplmxqmzbohnbhvaummzr ; /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941576.5659258-14514-42603695051684/AnsiballZ
_command.py",
        "root       42964  0.0  0.2  16948  2572 pts/2    Ss   14:52   0:00 sudo -H -S -n -u node /bin/sh -c echo BECOME-SUCCES
S-agvljtlylxktplmxqmzbohnbhvaummzr ; /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941576.5659258-14514-42603695051684/AnsiballZ
```

The log is longer because we called the new user ‘node’ and due to that there

are several processes with ‘node’ so ‘grep node’ brings them too

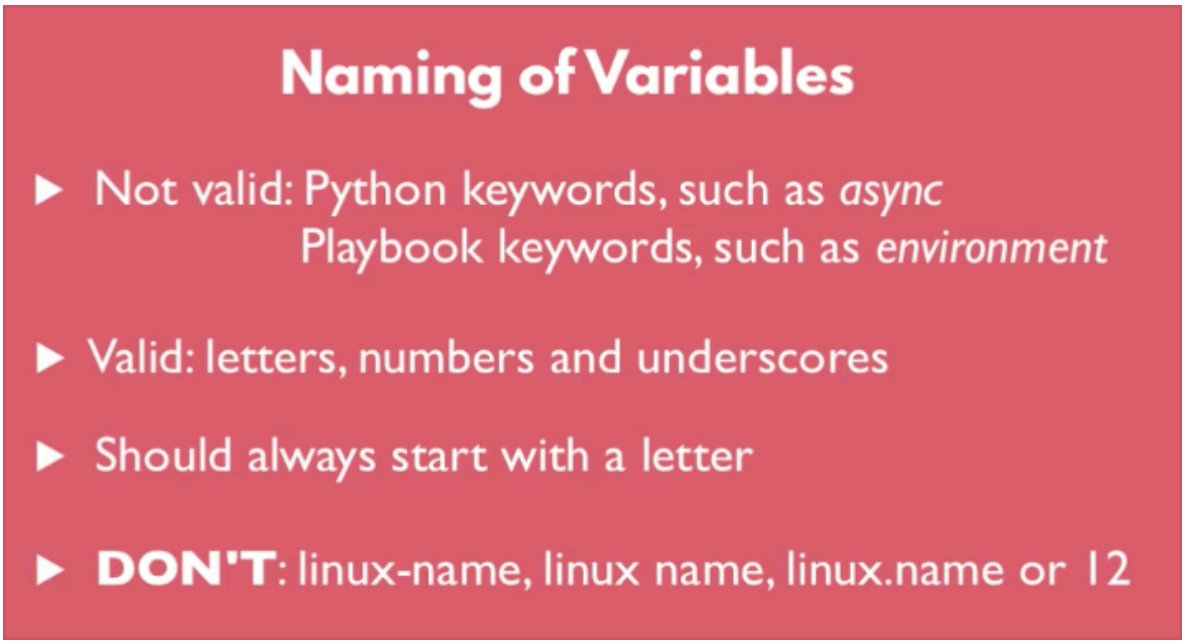
Here is the same result but from the droplets console

```
root@nodejs-app:~# ps aux | grep node
node        42935  0.0  2.1  40912 21072 ?        S    14:52   0:00 /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574.9672909-14502-50420489012334/async_wrapper.py j221932103497 1000 false /var/tmp/ansible-tmp-
1785941574.9672909-14502-50420489012334/AnsiballZ_command.py /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574.9672909-14502-50420489012334/AnsiballZ_command.py
node        42936  0.0  2.1  40912 21332 ?        S    14:52   0:00 /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574.9672909-14502-50420489012334/async_wrapper.py j221932103497 1000 false /var/tmp/ansible-tmp-
1785941574.9672909-14502-50420489012334/AnsiballZ_command.py /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574.9672909-14502-50420489012334/AnsiballZ_command.py
node        42937  0.9  2.9  40416 28584 ?        S    14:52   0:00 /usr/bin/python3.12 /var/tmp/ansible-tmp-1785941574.9672909-14502-50420489012334/AnsiballZ_command.py
node        42938  1.6  5.9 618312 58392 ?        Sl   14:52   0:00 node server
root        42977  0.0  0.2   7076  2168 pts/0    S+   14:53   0:00 grep --color=auto node
root@nodejs-app:~#
```

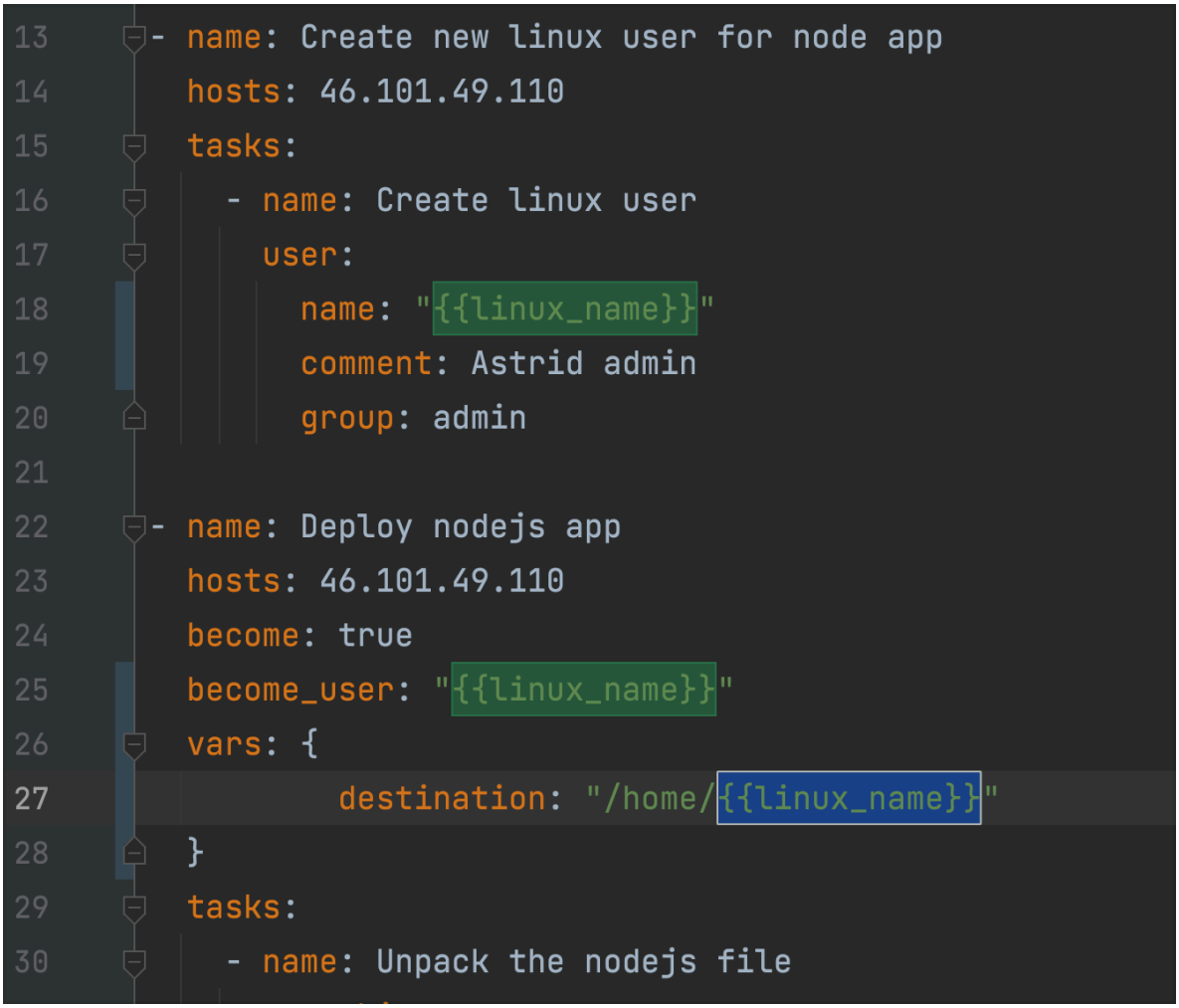
We also got this warning

```
) ansible-playbook -i hosts deploy-node.yaml -e "version=1.0.0 location=/Users/astrid/PythonProject/Ansible/nodejs-app name=nod
e"
[WARNING]: Found variable using reserved name 'name'.
Origin: <CLI option '-e'>

name


PLAY [Install node and npm] ****************************************************************************************************
```

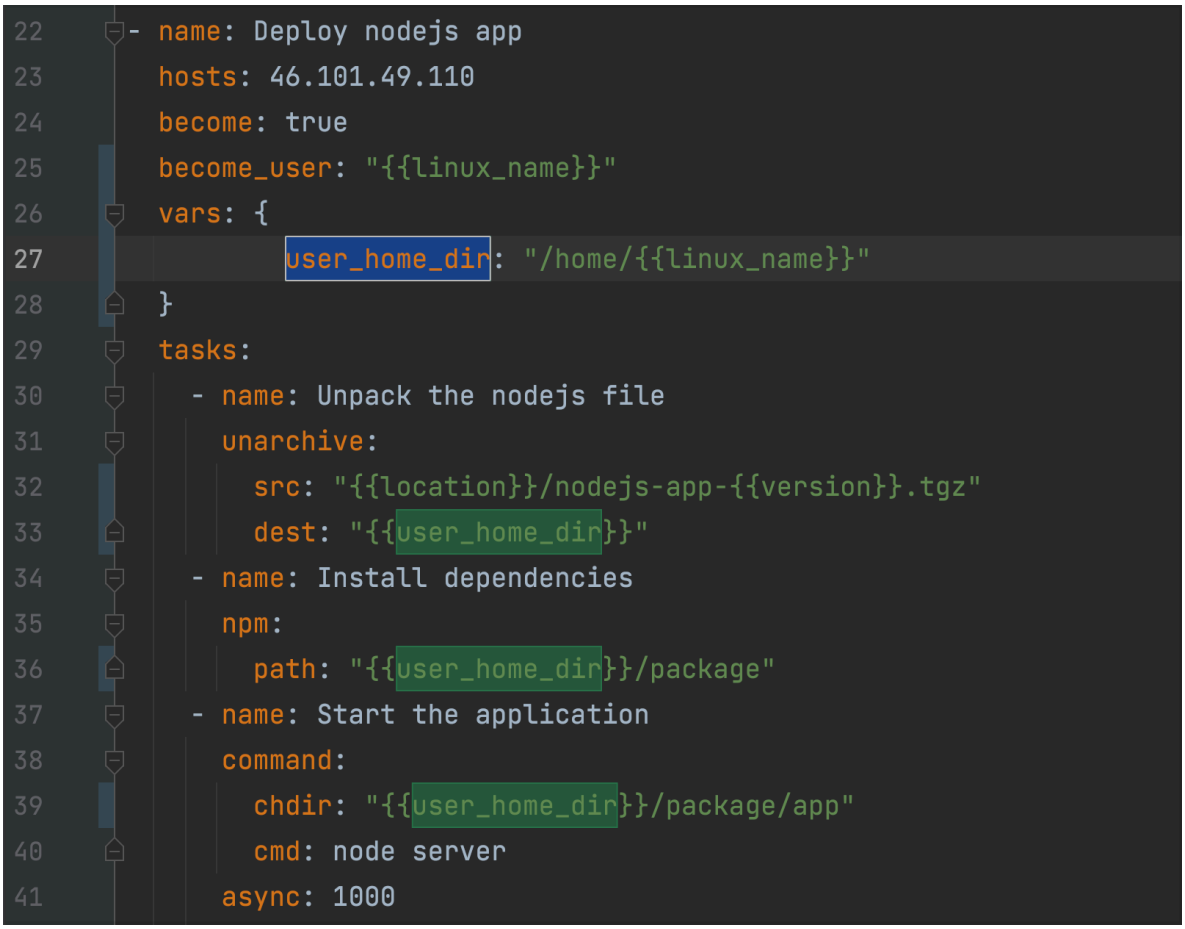
Regarding the use of a ‘reserved word’ and we should avoid doing that



Let’s refactor



Same for ‘destination’ it is too vague.



```
- name: Deploy nodejs app
  hosts: 46.101.49.110
  become: true
  become_user: "{{linux_name}}"
  vars: {
          user_home_dir: "/home/{{linux_name}}"
  }
  tasks:
    - name: Unpack the nodejs file
      unarchive:
        src: "{{location}}/nodejs-app-{{version}}.tgz"
        dest: "{{user_home_dir}}"
    - name: Install dependencies
      npm:
        path: "{{user_home_dir}}/package"
    - name: Start the application
      command:
        chdir: "{{user_home_dir}}/package/app"
        cmd: node server
      async: 1000
```

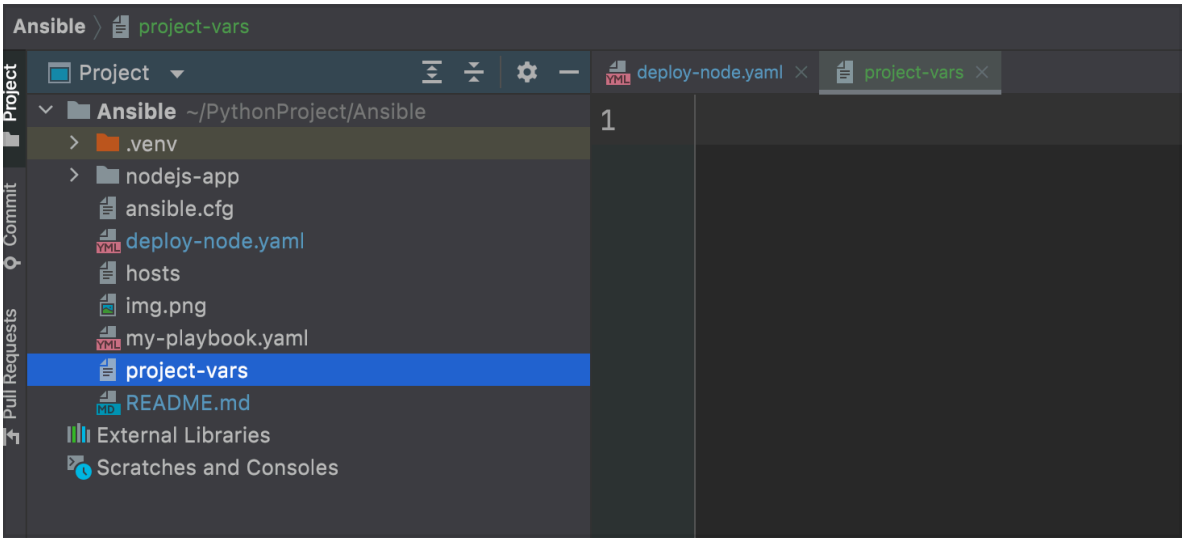
External Variables File

But if I have too many parameters it is not convinient to pass all those arguments in the terminal.

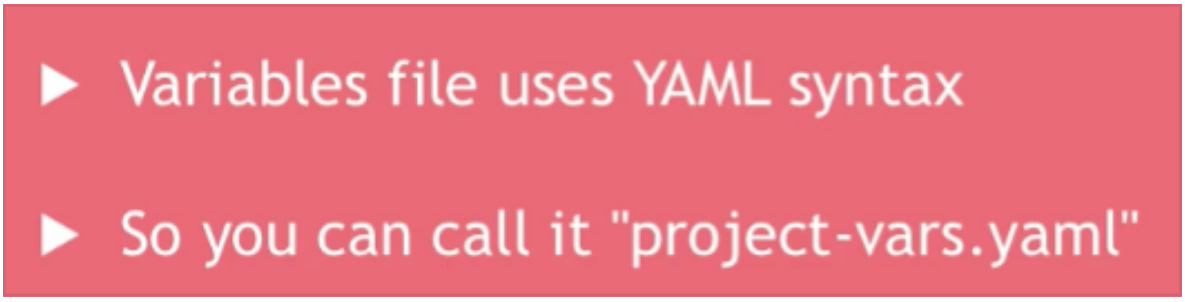
The solution Is to use a separate variables file

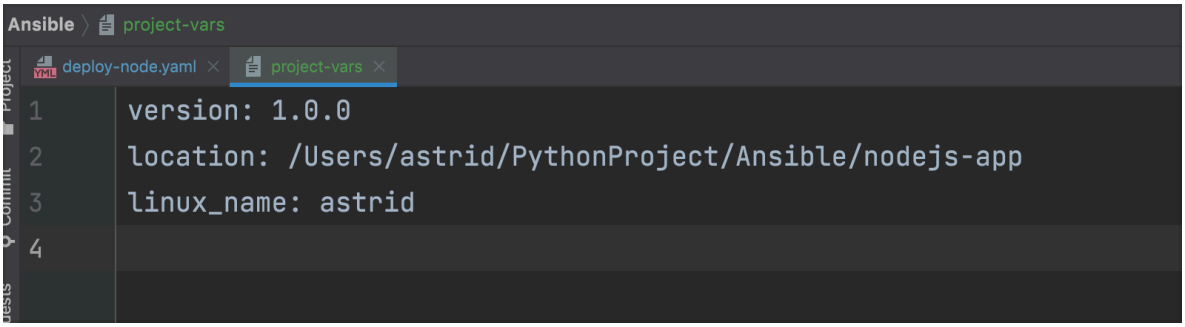
Then we will tell Ansible to use the vars file.

So I created a new text file (it can also be a yaml file) and I called it ‘project-vars’ or anything I want

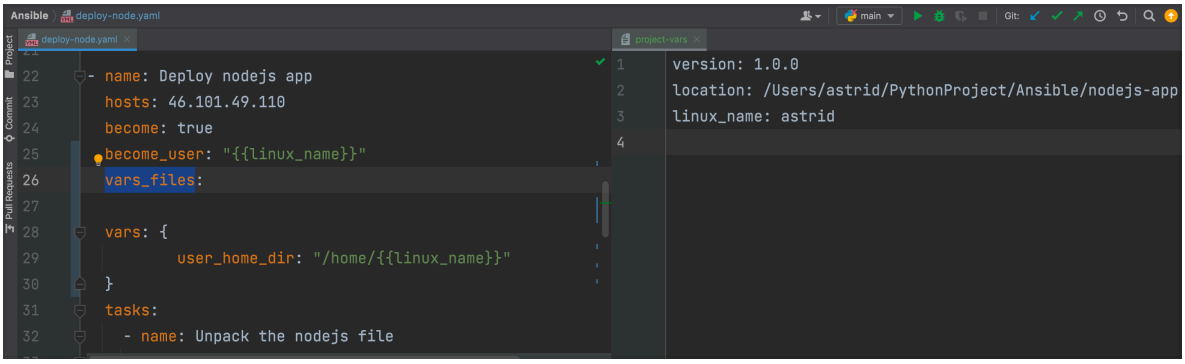


And then I write the variables and their values

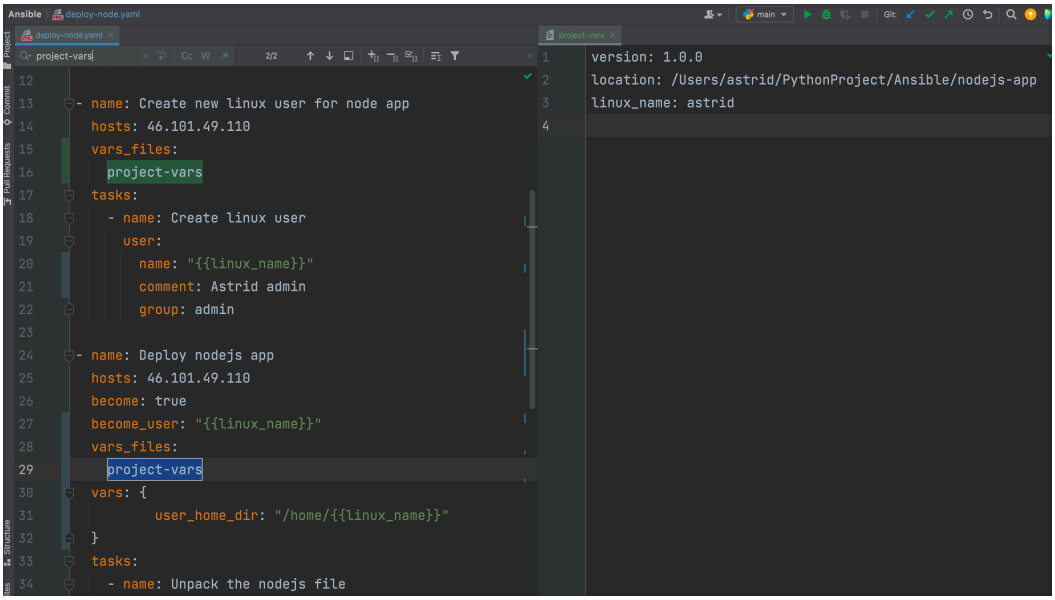




And then I reference the project-vars file inside the playbook using the attribute ‘vars_files’



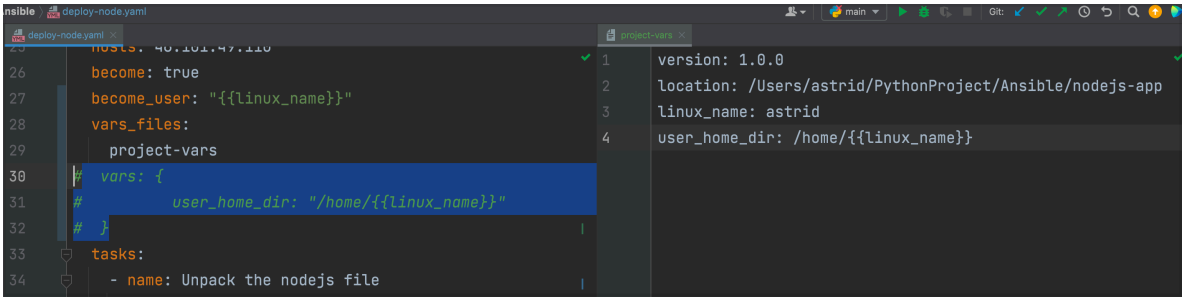
in each play I need it:



```
- name: Create new linux user for node app
  hosts: 46.101.49.110
  vars_files:
    project-vars
  tasks:
    - name: Create linux user
      user:
        name: "{{linux_name}}"
        comment: Astrid admin
        group: admin

- name: Deploy nodejs app
  hosts: 46.101.49.110
  become: true
  become_user: "{{linux_name}}"
  vars_files:
    project-vars
  vars: {
        user_home_dir: "/home/{{linux_name}}"
  }
  tasks:
    - name: Unpack the nodejs file
```

```
version: 1.0.0
location: /Users/astrid/PythonProject/Ansible/nodejs-app
linux_name: astrid
```

And we can also move the user_home_dir to the vars file



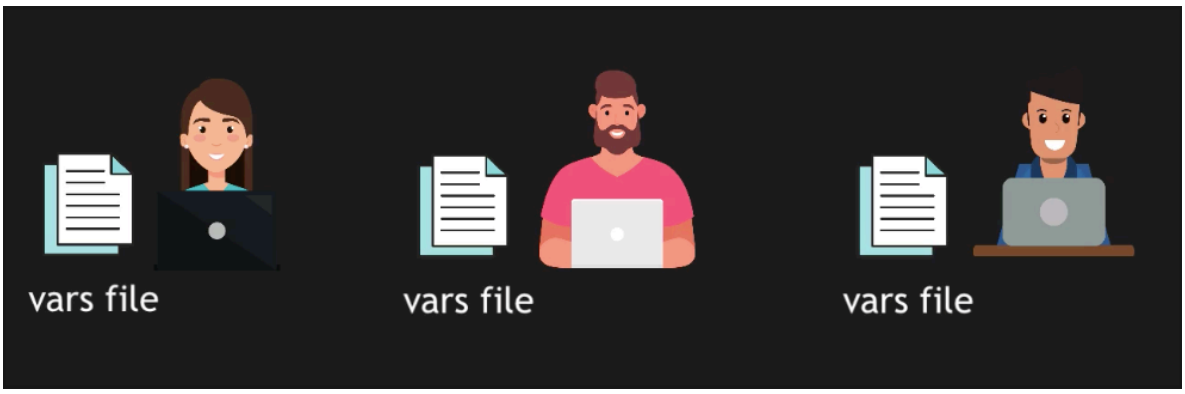
```
  become: true
  become_user: "{{linux_name}}"
  vars_files:
    project-vars
#  vars: {
#        user_home_dir: "/home/{{linux_name}}"
#  }
  tasks:
    - name: Unpack the nodejs file
```

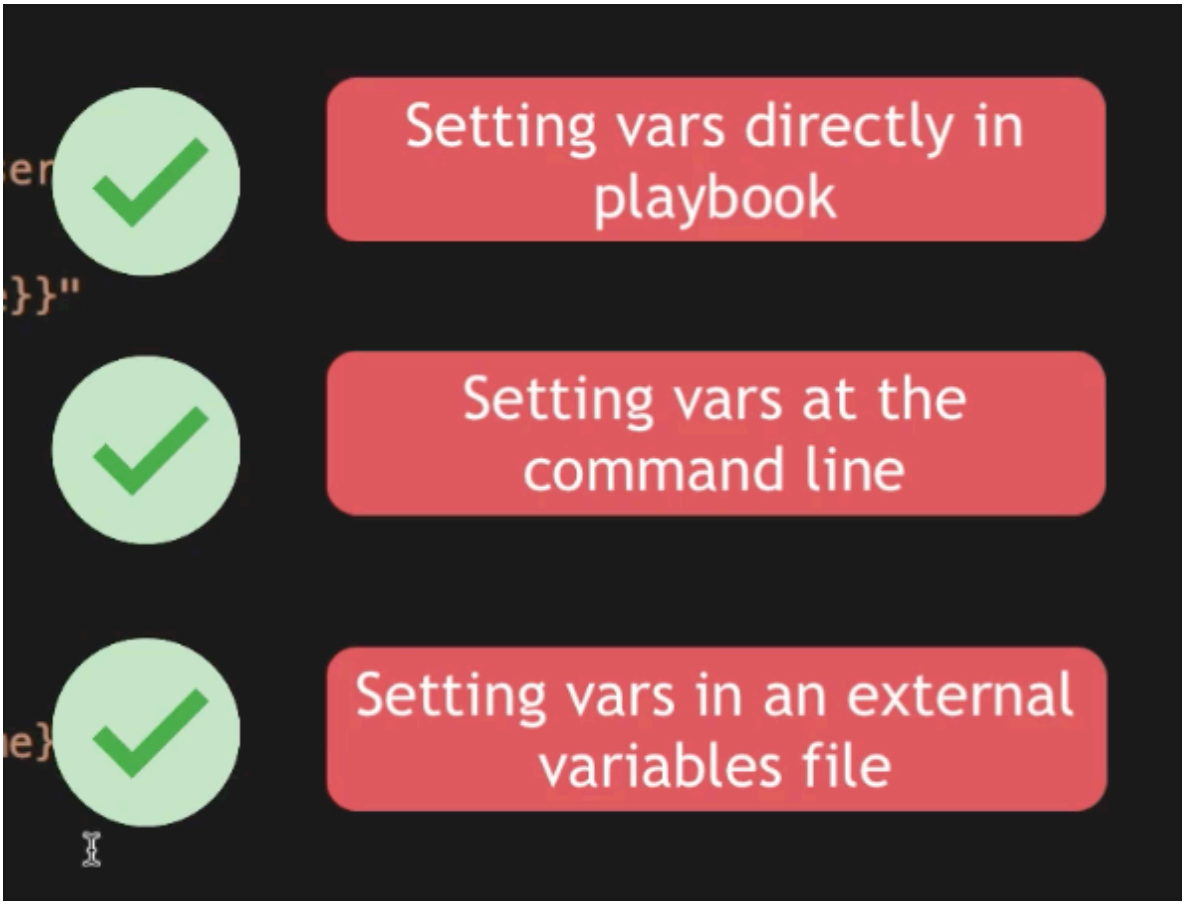
```
version: 1.0.0
location: /Users/astrid/PythonProject/Ansible/nodejs-app
linux_name: astrid
user_home_dir: /home/{{linux_name}}
```

And we execute the playbook again
-> ansible-playbook -i hosts deploy-node.yaml

```
TASK [Ensure app is running] ***************************************************************************************************
changed: [46.101.49.110]

TASK [debug] *******************************************************************************************************************
ok: [46.101.49.110] => {
    "msg": [
        "astrid     43686 53.5  6.4 618068 63072 ?        Sl   15:18   0:00 node server",
        "astrid     43706  0.0  0.1   2800  1920 pts/1    S+   15:18   0:00 /bin/sh -c ps aux | grep node",
        "astrid     43708  0.0  0.2   7076  2164 pts/1    S+   15:18   0:00 grep node"
    ]
}

PLAY RECAP *********************************************************************************************************************
46.101.49.110              : ok=11   changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

~/Pyt/Ansible  Module_15/Ansible_chapter-12 +2 !3 ?3                                                    ✔  20s  16:18:
```

Now every team member can have their own local version of the variables file:


vars file
vars file
vars file


Setting vars directly in playbook
Setting vars at the command line
Setting vars in an external variables file